EXPERIMENT 9

PAGE REPLACEMENT ALGORITHMS-

FIRST IN FIRST OUT(FIFO)

LEAST RECENTLY USED(LRU)

FIFO PAGE REPLAEMENT

```
#include <stdio.h>
int main(){
    int pages[20],frame[10],n,f,i,j,k=0,flag;
    scanf("%d",&n);
    for(i=0;i<n;i++) scanf("%d",&pages[i]);

    scanf("%d",&f);
    for(i=0;i<f;i++) frame[i]=-1;

    for(i=0;i<n;i++){
        flag=0;
        for(j=0;j<f;j++)
            if(frame[j]==pages[i]) flag=1;

        if(!flag){
            frame[k]=pages[i];
            k=(k+1)%f;
        }
    }
}
```

LEAST RECENTLY USED(LRU)

```
#include <stdio.h>
int main(){
    int pages[20],frame[10],time[10],n,f,i,j,pos,least;
    scanf("%d",&n);
    for(i=0;i<n;i++) scanf("%d",&pages[i]);

    scanf("%d",&f);
    for(i=0;i<f;i++) frame[i]=-1;

    for(i=0;i<n;i++){
        int found=0;
        for(j=0;j<f;j++){
            if(frame[j]==pages[i]){
                time[j]=i;
                found=1;
            }
        }

        if(!found){
            least=0;
            for(j=1;j<f;j++)
                if(time[j]<time[least]) least=j;

            frame[least]=pages[i];
            time[least]=i;
        }
    }
}
```

OUTPUT

Sample Input

```
Pages: 1 3 0 3 5 6 3
Frames = 3
```

FIFO (First In First Out)

Step	Page	Frames (3)	Page Fault
1	1	1 - -	Yes
2	3	1 3 -	Yes
3	0	1 3 0	Yes
4	3	1 3 0	No
5	5	5 3 0	Yes
6	6	5 6 0	Yes
7	3	5 6 3	Yes

Total Page Faults = 6

LRU (Least Recently Used)

Step	Page	Frames (3)	Page Fault
1	1	1 - -	Yes
2	3	1 3 -	Yes
3	0	1 3 0	Yes
4	3	1 3 0	No
5	5	5 3 0	Yes
6	6	5 3 6	Yes
7	3	5 3 6	No

Total Page Faults = 5